
PH102: Physics-2

Electrodynamics

Lecture # 05

Electrostatics-Electric field and Gauss's
Law

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Previous lecture

➤ Spherical polar coordinates

$$x = r \cos \theta \sin \varphi, \quad y = r \sin \theta \sin \varphi, \quad z = r \cos \theta$$

$$d\mathbf{l} = \hat{\mathbf{r}}dl_r + \hat{\boldsymbol{\theta}}dl_\theta + \hat{\boldsymbol{\phi}}dl_\phi = dr\hat{\mathbf{r}} + r d\theta\hat{\boldsymbol{\theta}} + r\sin\theta d\phi\hat{\boldsymbol{\phi}}$$

$$h_1 = 1, h_2 = r, h_3 = r\sin\theta$$

➤ Cylindrical polar coordinates

$$x = s \cos \varphi, \quad y = s \sin \varphi, \quad z = z$$

$$d\mathbf{l} = \hat{\mathbf{s}}dl_s + \hat{\boldsymbol{\phi}}dl_\phi + \hat{\mathbf{z}}dl_z = ds \hat{\mathbf{s}} + s d\phi \hat{\boldsymbol{\phi}} + dz \hat{\mathbf{z}}$$

$$h_1 = 1, h_2 = s, h_3 = 1$$

➤ Dirac delta function

$$\boldsymbol{\nabla} \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = 4\pi\delta^3(\mathbf{r})$$

Maxwell's equations

Our objective is to use and understand the significance of the gradient, divergence and curl of the scalar/vector fields for meaningful physical quantities. In the due course of time, we shall be in a position to express these quantities in the form of Maxwell's equations.

$$\left. \begin{aligned} \nabla \cdot \mathbf{E} &= \frac{1}{\epsilon_0} \rho \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \cdot \mathbf{B} &= 0 \\ \nabla \times \mathbf{B} &= \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \end{aligned} \right\} \begin{aligned} &\text{The Maxwell's equations in vacuum.} \\ &\text{These are space as well as} \\ &\text{time dependent in nature.} \end{aligned}$$

In order to understand the simple nature of these equations, we shall consider a case where fields in the equations are independent of time – **static case**. All charges are permanently fixed in space or, if they move, they do so as a **steady flow** in a circuit (ρ and $\mathbf{J}$ are constant in time).

(Contd...)

Static case:

- Fields are independent of time.
- Two sets of equations are obtained.
- The electric field appears in the first two equations.
- The magnetic field appears in the other two equations.
- The two fields are not interconnected to each other.

$$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \rho, \quad \nabla \times \mathbf{E} = 0$$

Electrostatics

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J}$$

Magnetostatics

Electricity and magnetism are distinct phenomena so long as the charges and currents are static in nature.

Electric charge

An electric charge is the essence of electrical phenomena. It manifest itself due to the forces of interaction.

Electric charge has the following characteristics:

- **Two kinds of electric charge.** There is a **positive** and a **negative** charge. The magnitude of the charge is given by a positive real number, and its type is denoted by a plus (+) or minus (-) sign. Two charges of the same sign physically repel each other, but two charges of opposite sign attract each other.
- **Principle of conservation of charge.** In nature, the total amount of positive charge just balances the total amount of negative charge; **electrical neutrality of objects is the most common occurrence.** Moreover, it is not possible to create (or annihilate) positive (or negative) charge without creating or annihilating an equal amount of negative (or positive) charge. This may be regarded as a principle of conservation of charge.

- **Charge exhibits symmetry.** We see "approximate" manifestations of this effect in the realm of elementary particle physics. For every elementary particle with a positive charge, there exists another "identical" elementary particle that has a negative charge of equal magnitude. Thus, we have electrons and positrons, protons and antiprotons, π^+ mesons and π^- mesons, and so on.
- **Charge is quantized.** This means that there seems to be a minimum magnitude (nonzero) to electric charge. This minimum magnitude is associated, for example, with the charge of a positron or electron. Therefore, all charges are integral multiples of this elementary charge.
- **Whatever has charge has mass.** Objects with zero (rest) mass have no charge.

For the electron (negative charge)
Rest mass = 9.109×10^{-31} kg.
Charge (e) = -1.602×10^{-19} C.

For the proton (positive charge)
Rest mass = 1.672×10^{-27} kg.
Charge (e) = $+1.602 \times 10^{-19}$ C.

Electric field lines

- Electric field lines provide a convenient graphical representation of the electric field in space.
- The field lines for a positive and a negative charges are shown in Fig. a & b.
- Notice that the direction of field lines is radially outward for a positive charge and radially inward for a negative charge.
- For a pair of charges of equal magnitude but opposite sign (an electric dipole), the field lines are shown in Fig. c.

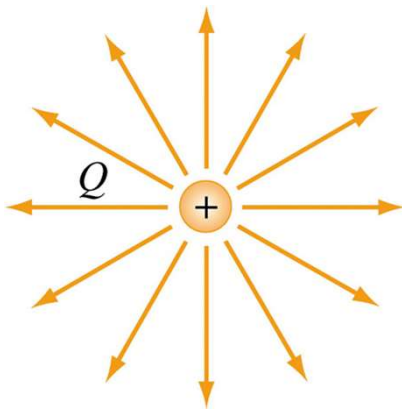


Fig a

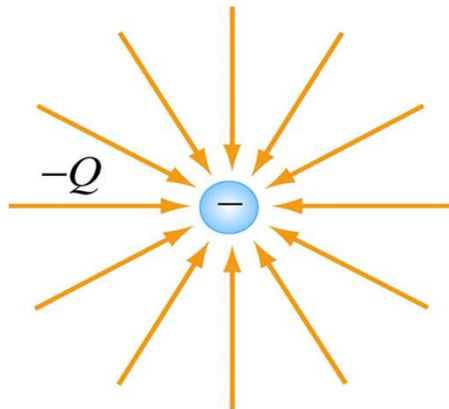


Fig b

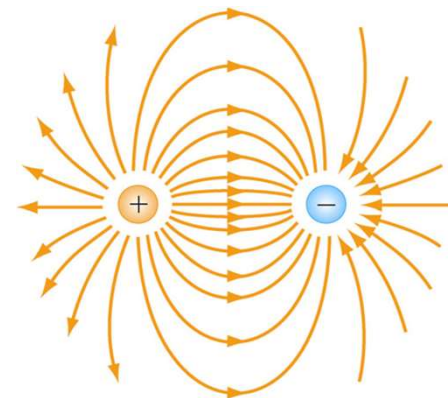


Fig c

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The pattern of electric field lines can be obtained by considering the following:

- **Symmetry:** For every point above the line joining the two charges there is an equivalent point below it. Therefore, the pattern must be symmetrical about the line joining the two charges (Fig. a).
- **Near field:** Very close to a charge, the field due to that charge predominates. Therefore, the lines are radial and spherically symmetric (Fig. a).
- **Far field:** Far from the system of charges, the pattern should look like that of a single point charge of value $Q = \sum_i Q_i$. Thus, the lines should be radially outward, unless $Q = 0$ (Fig. a).
- **Null point:** This is a point where $\vec{E} = 0$, and no field lines should pass through it (Fig. b).

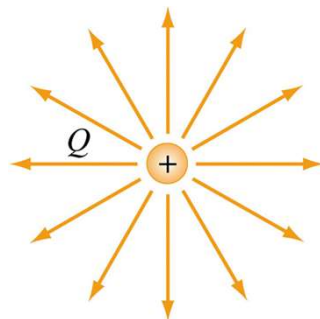


Fig. a

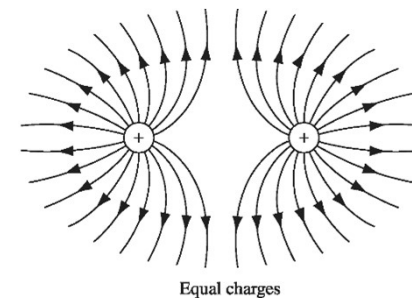


Fig. b

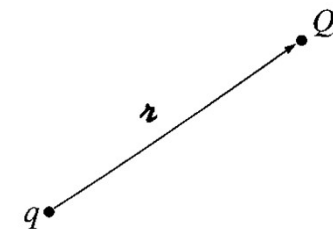
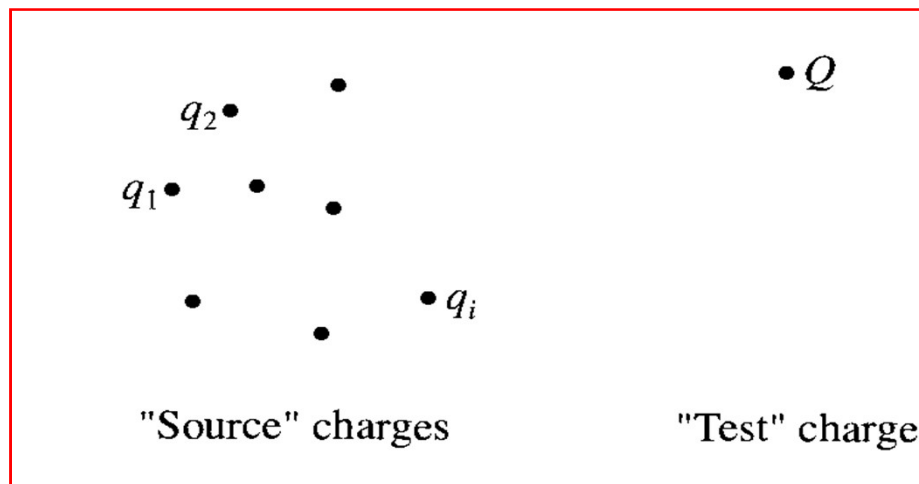
Properties of the electric field lines

- ✓ The direction of the electric field vector at a point is tangent to the field lines.
- ✓ The number of lines per unit area through a surface perpendicular to the line is proportional to the magnitude of the electric field in a given region.
- ✓ The field lines must begin on positive charges (or at infinity) and then terminate on negative charges (or at infinity).
- ✓ The number of lines that originate from a positive charge or terminating on a negative charge must be proportional to the magnitude of the charge.
- ✓ No two field lines can cross each other; otherwise the field would be pointing in two different directions at the same point which is meaningless.

The Electric Field: *Introduction*

In electrostatics, we are concerned with the force acting on a test charge Q due to a source charge q .

- In general, both the source charges and the test charge are in motion.
- We shall consider the special case of the **electrostatics** in which all the source charges are stationary.
- The **principle of superposition** states that the interaction between any two charges is completely unaffected by the presence of others.



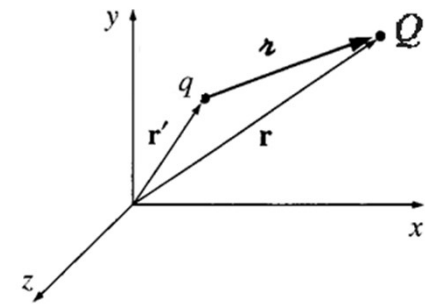
Interaction of one of the isolated charge.

Coulomb's law

- To determine the force on Q , we can first compute the force $\mathbf{F}_1$, due to q_1 alone (ignoring all the others); then we compute the force $\mathbf{F}_2$, due to q_2 alone; and so on.
- The net force is the vector sum of all these individual forces: $\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \dots$

The force on a test charge Q due to a single point charge q which is at rest a distance r away is given by

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \hat{\mathbf{r}}.$$



$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$ (the permittivity of free space).

$\mathbf{r} = \mathbf{r} - \mathbf{r}'$; is the separation vector from $\mathbf{r}'$ (the location of q) to $\mathbf{r}$ (the location of Q)

Coulomb's law is based on the experimental results.

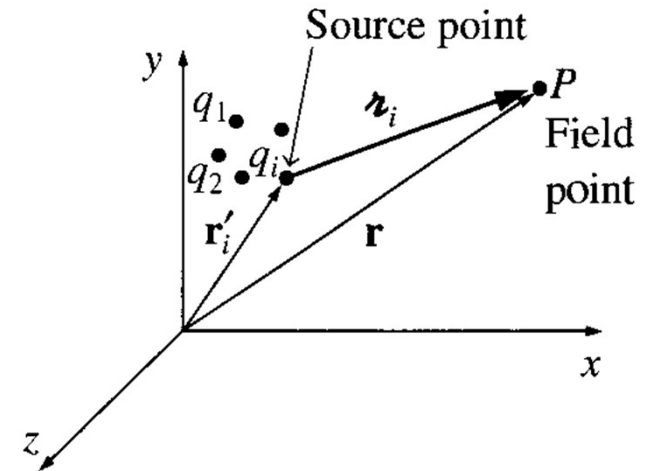
How does one particle sense the presence of the other?

- The electric charge creates an electric field in the space around it.
- A second charged particle does not interact directly with the first; rather, it responds to whatever field it encounters. In this sense, the field acts as an *intermediary* between the particles.
- If we have several point charges $q_1, q_2, \dots, q_n$ at distances $r_1, r_2, \dots, r_n$ from Q , the total force on Q is

$$\begin{aligned}\mathbf{F} &= \mathbf{F}_1 + \mathbf{F}_2 + \dots = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 Q}{r_1^2} \hat{\mathbf{r}}_1 + \frac{q_2 Q}{r_2^2} \hat{\mathbf{r}}_2 + \dots \right) \\ &= \frac{Q}{4\pi\epsilon_0} \left(\frac{q_1}{r_1^2} \hat{\mathbf{r}}_1 + \frac{q_2}{r_2^2} \hat{\mathbf{r}}_2 + \frac{q_3}{r_3^2} \hat{\mathbf{r}}_3 + \dots \right), \\ \mathbf{F} &= QE, \quad \mathbf{E}(\mathbf{r}) \equiv \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i^2} \hat{\mathbf{r}}_i.\end{aligned}$$

(Contd...)

$\mathbf{E}$ is called the electric field of the source charges. It is a function of position ($\mathbf{r}$), because the separation vectors $\mathbf{r}_i$ depends on the location of the field point P . But it makes no reference to the test charge Q .



The electric field is a vector quantity that varies from point to point and is **determined by the configuration of source charges**; physically, $\mathbf{E}(\mathbf{r})$ is the force per unit charge that would be exerted on a test charge, if you were to place one at P .

Equation of line of force

If there is a curve in space, $\mathbf{r}(s)$, whose tangent vector, $\hat{\mathbf{t}}(\mathbf{r})$ at every point lies parallel to the electric field at that point, we have what is referred to as a line of force (or electric field line). If displacements $\mathbf{r}_1$ and $\mathbf{r}$ locate two points on the curve and $\Delta\mathbf{r} = \mathbf{r}_1 - \mathbf{r}$, then the necessary condition for the curve to represent a line of force is that, as $\mathbf{r}_1$ approaches $\mathbf{r}$,

$$\lim_{\Delta\mathbf{r} \rightarrow 0} \frac{\Delta\mathbf{r}}{\Delta s} = \frac{d\mathbf{r}}{ds} \equiv \hat{\mathbf{t}}(\mathbf{r}) = \alpha \mathbf{E}(\mathbf{r})$$

where α is a scalar constant and s is taken to measure distance along the curve from some arbitrary point on the curve. If $\mathbf{r} = x\hat{x} + y\hat{y} + z\hat{z}$, then

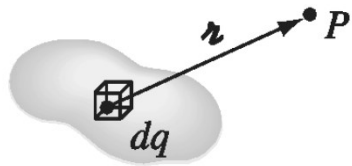
$$\frac{1}{\alpha} \frac{dx}{ds} = E_x, \quad \frac{1}{\alpha} \frac{dy}{ds} = E_y, \quad \frac{1}{\alpha} \frac{dz}{ds} = E_z$$

$$\frac{dx}{dE_x} = \frac{dy}{dE_y} = \frac{dz}{dE_z}$$

Continuous Charge Distributions

Divide the charge distribution into infinitesimal elements of charge dq which may be considered to be point charges.

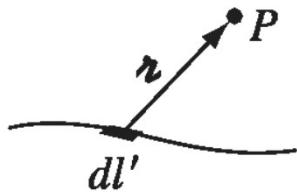
The charge is distributed continuously over some region, the sum becomes an integral.



$$d\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2} \hat{\mathbf{r}} \Rightarrow \mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r^2} \hat{\mathbf{r}}$$

The charge is spread out along a line, with charge-per-unit-length λ , then $dq = \lambda dl'$

The electric field of a line charge is

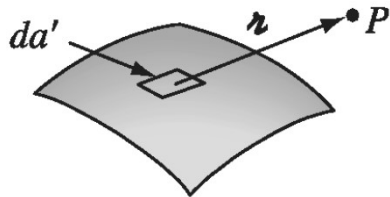


$$\mathbf{E}(r) = \frac{1}{4\pi\epsilon_0} \int \frac{\lambda(r')}{r^2} \hat{\mathbf{r}} dl'$$

Read r as $\mathbf{r}$

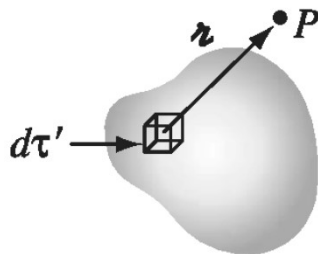
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The charge is smeared out over a surface, with charge-per-unit-area σ , then $dq = \sigma da'$ (where da' is an element of area on the surface). The electric field for a surface charge is



$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma(\mathbf{r}')}{r^2} \hat{\mathbf{r}} da'$$

The charge fills a volume, with charge-per-unit-volume ρ , then $dq = \rho d\tau'$ (where $d\tau'$ is an element of volume). The electric field for a volume charge is



$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{r^2} \hat{\mathbf{r}} d\tau'$$

Read r as $\hat{r}$

(Contd...)

	Line charge	Ring of charge	Uniformly charged disk
Figure			
(2) Express dq in terms of charge density	$dq = \lambda dx'$	$dq = \lambda d\ell$	$dq = \sigma dA$
(3) Write down dE	$dE = k_e \frac{\lambda dx'}{r'^2}$	$dE = k_e \frac{\lambda d\ell}{r^2}$	$dE = k_e \frac{\sigma dA}{r^2}$
(4) Rewrite r and the differential element in terms of the appropriate coordinates	dx' $\cos \theta = \frac{y}{r'}$ $r' = \sqrt{x'^2 + y^2}$	$d\ell = R d\phi'$ $\cos \theta = \frac{z}{r}$ $r = \sqrt{R^2 + z^2}$	$dA = 2\pi r' dr'$ $\cos \theta = \frac{z}{r}$ $r = \sqrt{r'^2 + z^2}$
(5) Apply symmetry argument to identify non-vanishing component(s) of dE	$dE_y = dE \cos \theta$ $= k_e \frac{\lambda y dx'}{(x'^2 + y^2)^{3/2}}$	$dE_z = dE \cos \theta$ $= k_e \frac{\lambda R z d\phi'}{(R^2 + z^2)^{3/2}}$	$dE_z = dE \cos \theta$ $= k_e \frac{2\pi \sigma z r' dr'}{(r'^2 + z^2)^{3/2}}$
(6) Integrate to get E	$E_y = k_e \lambda y \int_{-\ell/2}^{+\ell/2} \frac{dx'}{(x'^2 + y^2)^{3/2}}$ $= \frac{2k_e \lambda}{y} \frac{\ell/2}{\sqrt{(\ell/2)^2 + y^2}}$	$E_z = k_e \frac{R \lambda z}{(R^2 + z^2)^{3/2}} \oint d\phi'$ $= k_e \frac{(2\pi R \lambda) z}{(R^2 + z^2)^{3/2}}$ $= k_e \frac{Qz}{(R^2 + z^2)^{3/2}}$	$E_z = 2\pi \sigma k_e z \int_0^R \frac{r' dr'}{(r'^2 + z^2)^{3/2}}$ $= 2\pi \sigma k_e \left(\frac{z}{ z } - \frac{z}{\sqrt{z^2 + R^2}} \right)$

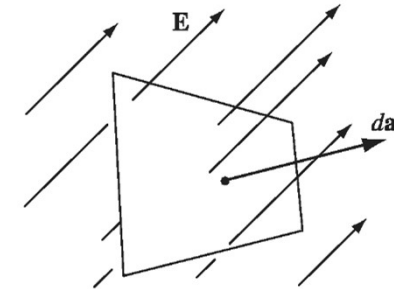
Gauss's law (integral form)

The strength of an electric field is **proportional to the number of field lines per unit area**. The number of electric field lines that penetrates a given surface is called an **electric flux**, which we denote as Φ_E .

The flux through surface S is a measure of number of field lines passing through S .

$$\Phi_E = \int_S \mathbf{E} \cdot d\mathbf{a}$$

The dot product picks out the component of $d\mathbf{a}$ along the direction of $\mathbf{E}$, as indicated in the figure.

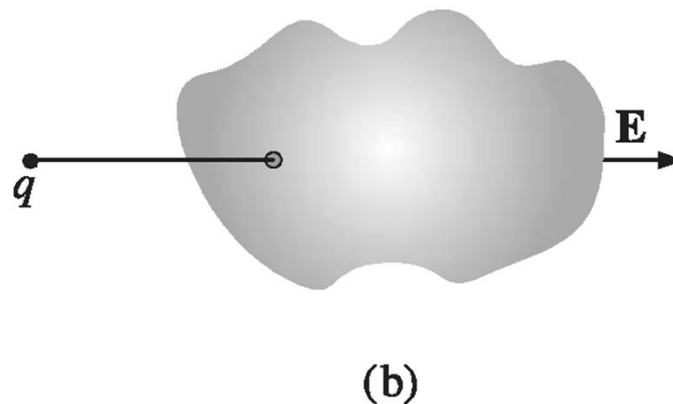
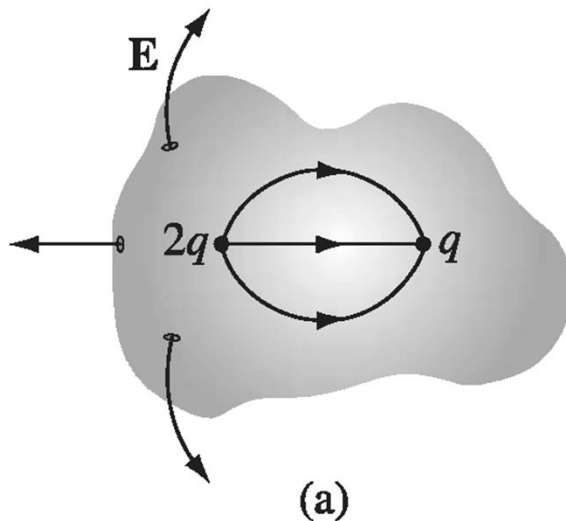


The field-line diagram on a two-dimensional surface is deceptive. The density of lines passing through a circle of radius r is the total number divided by the circumference ($n/2\pi r$), which goes like $(1/r)$, not $(1/r^2)$.

In the model in three dimensions, the density of lines is the total number divided by the area of the sphere ($n/4\pi r^2$), which *does* go like $(1/r^2)$.

(Contd...)

- The flux through any closed surface is a measure of the total charge inside. For the field lines that originate on a positive charge must either pass out through the surface or else terminate on a negative charge inside (Fig. a).
- On the other hand, a charge outside the surface will contribute nothing to the total flux, since its field lines pass in one side and out the other (Fig. b). This is the essence of Gauss's law.



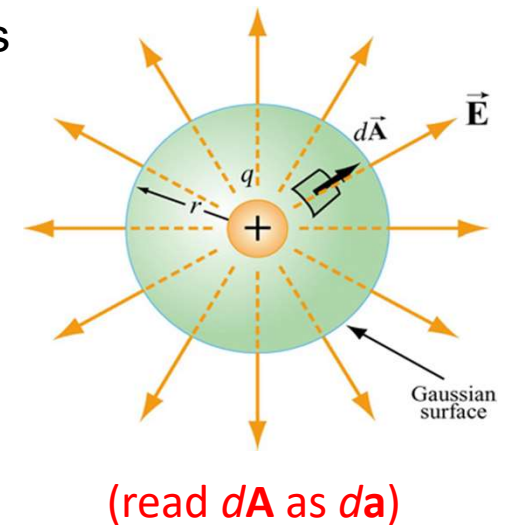
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For a single positive point charge q located at the center of a sphere of radius r , as shown in the figure, the electric field due to the charge q is

$$\mathbf{E} = \left(\frac{q}{4\pi\epsilon_0 r^2} \right) \hat{r}$$

The flux of $\mathbf{E}$ through a spherical surface of radius r is

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \int_S \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r^2} \hat{r} \right) \cdot (r^2 \sin\theta d\theta d\phi \hat{r}) = \frac{q}{\epsilon_0}$$



Notice that it all hinges on the $1/r^2$ character of Coulomb's law; without that the crucial cancellation of the r 's in in the above equation would not take place, and the total flux of $\mathbf{E}$ would depend on the surface chosen, not merely on the total charge enclosed.

For a collection of charges, according to the superposition principle, the total field is the vector sum of all the individual fields.

$$\mathbf{E} = \sum_{i=1}^n \mathbf{E}_i$$

The flux through a surface that encloses them all is

$$\oint \mathbf{E} \cdot d\mathbf{a} = \sum_{i=1}^n \left(\oint \mathbf{E} \cdot d\mathbf{a} \right) = \sum_{i=1}^n \left(\frac{q_i}{\epsilon_0} \right)$$

For any closed surface,

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{enc}$$

where Q_{enc} is the total charge enclosed within the surface.

The Gauss's law states that the net flux through any closed surface is proportional to the net charge enclosed.

Gauss's law (differential form)

We can convert Gauss law in to differential equation by applying the divergence theorem

The total charge enclosed Q_{enc} can be written as

Gauss's law becomes

So, Gauss's law in differential form

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \int_v (\nabla \cdot \mathbf{E}) d\tau$$

$$Q_{enc} = \int_v \rho(\mathbf{r}) d\tau$$

$$\int_v (\nabla \cdot \mathbf{E}) d\tau = \int_v \frac{\rho(\mathbf{r})}{\epsilon_0} d\tau$$

$$(\nabla \cdot \mathbf{E}) = \frac{\rho(\mathbf{r})}{\epsilon_0} \text{ (Maxwell's first equation)}$$

The divergence of $\mathbf{E}$ can be calculated as (using expression for $\mathbf{E}$ in terms of ρ)

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_{\text{all space}} \frac{\hat{\mathbf{r}}}{r^2} \rho(\mathbf{r}') d\tau'. \quad \nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = 4\pi\delta^3(\mathbf{r}).$$

$$\nabla \cdot \mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) \rho(\mathbf{r}') d\tau'.$$

$$\nabla \cdot \mathbf{E} = \frac{1}{4\pi\epsilon_0} \int 4\pi\delta^3(\mathbf{r} - \mathbf{r}') \rho(\mathbf{r}') d\tau' = \frac{1}{\epsilon_0} \rho(\mathbf{r})$$

Gauss's law in differential form

(Contd...)

Gauss's law provides a convenient tool for evaluating electric field. Its application is limited only to systems that possess certain symmetry, as listed below.

Symmetry	System	Gaussian surface
Planar	Infinite plane	Gaussian pill box
Cylindrical	Infinite rod	Coaxial cylinder
Spherical	Sphere, spherical shell	Concentric sphere

The following steps may be useful when applying Gauss's law:

1. Identify the symmetry associated with the charge distribution.
2. Determine the direction of the electric field, and a "Gaussian surface" on which the magnitude of the electric field is constant.
3. Calculate the electric flux Φ_E through the Gaussian surface.
4. Equate Φ_E with Q_{enc}/ϵ_0 and deduce the magnitude of the electric field.
5. In some cases, the space has to be divided into different regions associated with the charge distribution before carrying out the above calculation.

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System	Infinite line of charge	Infinite plane of charge	Uniformly charged solid sphere
Figure			
Identify the symmetry	Cylindrical	Planar	Spherical
Determine the direction of $\vec{E}$			
Divide the space into different regions	$r > 0$	$z > 0$ and $z < 0$	$r \leq a$ and $r \geq a$
Choose Gaussian surface			
Calculate electric flux	$\Phi_E = E(2\pi r l)$	$\Phi_E = EA + EA = 2EA$	$\Phi_E = E(4\pi r^2)$
Calculate enclosed charge q_{in}	$q_{enc} = \lambda l$	$q_{enc} = \sigma A$	$q_{enc} = \begin{cases} Q(r/a)^3 & r \leq a \\ Q & r \geq a \end{cases}$
Apply Gauss's law $\Phi_E = q_{in} / \epsilon_0$ to find E	$E = \frac{\lambda}{2\pi\epsilon_0 r}$ IITG	$E = \frac{\sigma}{2\epsilon_0}$ PH102	$E = \begin{cases} \frac{Qr}{4\pi\epsilon_0 a^3}, & r \leq a \\ \frac{Q}{4\pi\epsilon_0 r^2}, & r \geq a \end{cases}$ 2024

The curl of E

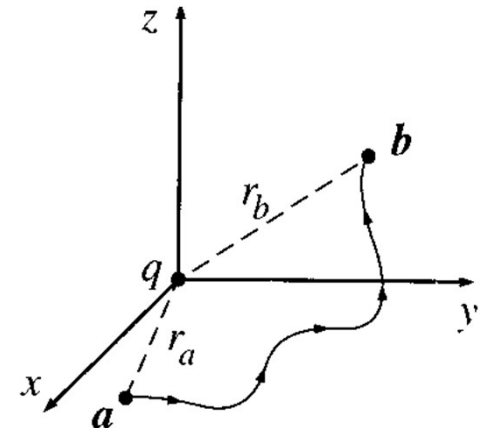
The electric field of a point charge located at the origin is $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$

The line integral of this field from some point **a** to some other point **b**, as shown in figure, is

$$\int_a^b \mathbf{E} \cdot d\mathbf{l}$$

In spherical coordinates, $d\mathbf{l} = dr\hat{r} + r d\theta\hat{\theta} + r \sin\theta d\phi\hat{\phi}$

$$\int_a^b \mathbf{E} \cdot d\mathbf{l} = \frac{1}{4\pi\epsilon_0} \int_a^b \frac{q}{r^2} dr = \left. \frac{-1}{4\pi\epsilon_0} \frac{q}{r} \right|_{r_a}^{r_b} = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r_a} - \frac{q}{r_b} \right)$$



Here, r_a is the distance from the origin to the point a and r_b is the distance to b.

The integral around a closed path is zero (because $r_a = r_b$) i.e., $\oint \mathbf{E} \cdot d\mathbf{l} = 0$

Therefore, by applying the Stokes' theorem,

$$\nabla \times \mathbf{E} = 0$$

The result holds no matter where the charge is located (coordinate system) and for any static charge distribution whatever.

Solved example 1

Find the lines of force of a point charge in the x-y plane.

The electric field due to a point charge is

$$\mathbf{E} = \left(\frac{q}{4\pi\epsilon_0 r^2} \right) \hat{r} = \left(\frac{q}{4\pi\epsilon_0} \right) \frac{x\hat{x} + y\hat{y}}{(x^2 + y^2)^{3/2}}$$

Substituting for E_x and E_y in

$$\frac{dx}{dE_x} = \frac{dy}{dE_y} = \frac{dz}{dE_z}$$

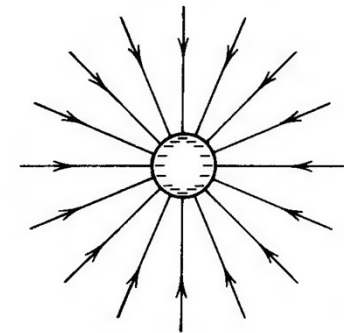
we get,

$$\frac{dy}{dx} = \frac{E_y}{E_x} = \frac{y}{x}$$

or,

$$y = kx$$

i.e., the equations of straight lines passing through the origin.
 k is a constant.



Solved example 2

Find the electric field (magnitude and direction) a distance z above the midpoint between two equal charges, q , a distance d apart as shown in figure a.

Let $\mathbf{E}_1$ be the field of the left charge alone, and $\mathbf{E}_2$ that of the right charge alone (fig b).

The horizontal components cancel each other while and the vertical components add each other.

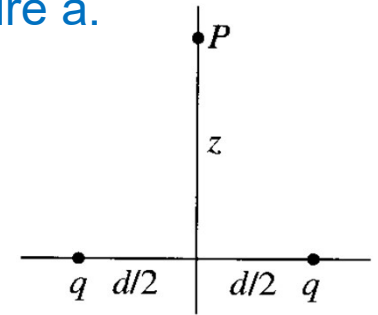


Fig a

$$E_z = 2 \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \cos \theta.$$

Here $r = \sqrt{z^2 + (d/2)^2}$ and $\cos \theta = z/r$, so

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{2qz}{[z^2 + (d/2)^2]^{3/2}} \hat{\mathbf{z}}.$$

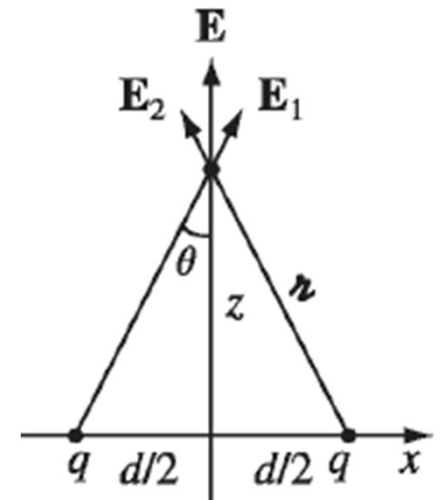
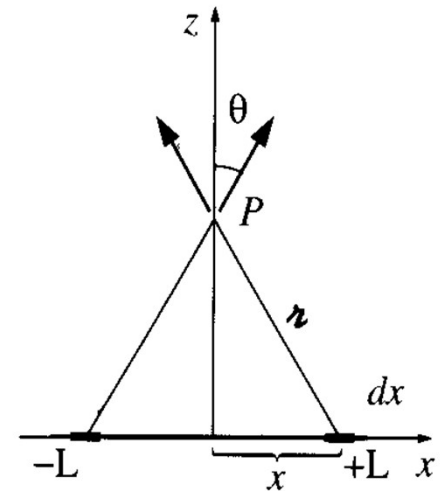


Fig b

Solved example 3

Find the electric field a distance z above the midpoint of a straight line segment of length $2L$, which carries a uniform line charge λ (Fig. below).

$$\begin{aligned}\mathbf{r} &= z \hat{\mathbf{z}}, \quad \mathbf{r}' = x \hat{\mathbf{x}}, \quad dl' = dx; \\ \mathbf{r} &= \mathbf{r} - \mathbf{r}' = z \hat{\mathbf{z}} - x \hat{\mathbf{x}}, \quad r = \sqrt{z^2 + x^2}, \quad \hat{\mathbf{r}} = \frac{\mathbf{r}}{r} = \frac{z \hat{\mathbf{z}} - x \hat{\mathbf{x}}}{\sqrt{z^2 + x^2}}. \\ \mathbf{E} &= \frac{1}{4\pi\epsilon_0} \int_{-L}^L \frac{\lambda}{z^2 + x^2} \frac{z \hat{\mathbf{z}} - x \hat{\mathbf{x}}}{\sqrt{z^2 + x^2}} dx \\ &= \frac{\lambda}{4\pi\epsilon_0} \left[z \hat{\mathbf{z}} \int_{-L}^L \frac{1}{(z^2 + x^2)^{3/2}} dx - \hat{\mathbf{x}} \int_{-L}^L \frac{x}{(z^2 + x^2)^{3/2}} dx \right] \\ &= \frac{\lambda}{4\pi\epsilon_0} \left[z \hat{\mathbf{z}} \left(\frac{x}{z^2 \sqrt{z^2 + x^2}} \right) \Big|_{-L}^L - \hat{\mathbf{x}} \left(-\frac{1}{\sqrt{z^2 + x^2}} \right) \Big|_{-L}^L \right] \\ &= \frac{1}{4\pi\epsilon_0} \frac{2\lambda L}{z \sqrt{z^2 + L^2}} \hat{\mathbf{z}}.\end{aligned}$$



From far away the line looks like a point charge $q = 2\lambda L$, so the field reduces to

$$\mathbf{E} = \frac{q}{4\pi\epsilon_0 z^2} \hat{\mathbf{z}}$$

Solved example 4

Find the field outside a uniformly charged solid sphere of radius R and total charge q .

Imagine a spherical surface at radius at $r > R$ which is called the Gaussian surface.

Gauss's law $\int_S \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{enc}$ Here, $Q_{enc} = q$

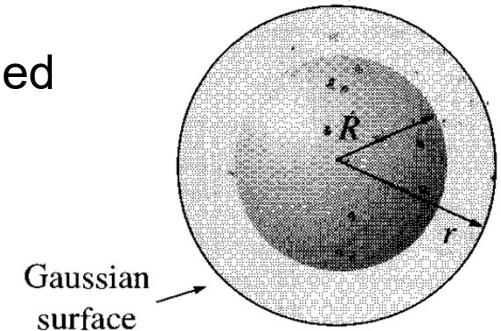
The spherical symmetry allows us to extract $\mathbf{E}$ from under integral sign since $\mathbf{E}$ points radially outward.

$$\int_S \mathbf{E} \cdot d\mathbf{a} = \int_S |\mathbf{E}| da$$

The magnitude of $\mathbf{E}$ is constant over the Gaussian surface, so it comes outside the integral.

$$\begin{aligned} \int_S |\mathbf{E}| da &= |\mathbf{E}| \int_S da = |\mathbf{E}| 4\pi r^2 \\ |\mathbf{E}| 4\pi r^2 &= \frac{q}{\epsilon_0}, \mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \end{aligned}$$

The field outside the sphere is exactly the same as it would have been if all the charge had been concentrated at the center.



Solved example 5

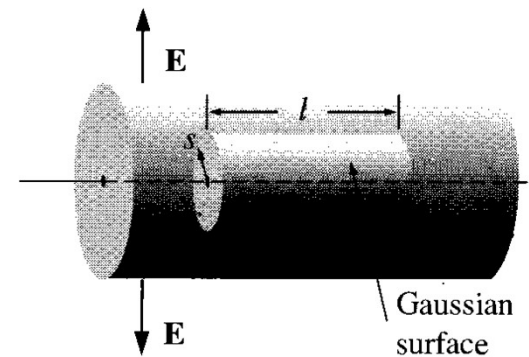
A long cylinder carries a charge density that is proportional to the distance from the axis: $\rho = ks$, for some constant k . Find the electric field inside this cylinder.

Draw a Gaussian cylinder of length l and radius R . The Gauss's law states that

$$\int_S \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{enc}$$

The charge enclosed is

$$Q_{enc} = \int \rho d\tau = \int (ks')s' ds' d\varphi dz = 2\pi kl \int_0^s s'^2 ds' = \frac{2}{3} \pi kls^3$$



The cylindrical symmetry dictates that $\mathbf{E}$ must point radially outward.

$$\int \mathbf{E} \cdot d\mathbf{a} = |\mathbf{E}| \int da = |\mathbf{E}| 2\pi ls$$

$$|\mathbf{E}| 2\pi ls = \frac{1}{\epsilon_0} \frac{2}{3} \pi kls^3$$

$$\mathbf{E} = \frac{1}{3\epsilon_0} ks^2 \hat{s}$$

Questions?